

Learning impurity incorporation energetics across host and impurity chemistry in two-dimensional materials with a defect-aware graph Transformer

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(Dated: August 10, 2026)

We study retrospective estimation and reranking of impurity formation energies for available DFT-relaxed candidate geometries in two-dimensional materials. In IMP2D, each relaxed geometry and its reference energy are products of the same completed DFT workflow; this task therefore evaluates database-conditioned prediction rather than replacing candidate generation or relaxation. Random structure splits nevertheless conceal how errors change with chemical novelty. We develop the Defect-Aware Radial Transformer (DART) and evaluate it on a curated set of 10,224 neutral structures from the Impurities in Two-Dimensional Materials (IMP2D) database. Across five paired repeats, a 2^3 factorial showed small error reductions from gated graph readout and the pre-normalized, distance-gated local block, whereas the independent effect of explicit environment enrichment remained inconclusive. For pair-held-out structures, all but one test structure had both its host and impurity represented elsewhere in training; the mean absolute error was 0.450 eV, compared with 0.418 eV under random cross-validation. Withholding every structure of a host or impurity increased the error to 0.938 eV and 0.844 eV, respectively. Incorporation geometry formed another applicability axis: pair-held-out interstitial structures had 0.719 eV error, more than twice the 0.322 eV error for adsorbates. For candidates sharing a host and impurity, pair-held-out predictions selected the lower-energy incorporation class with 91.0% accuracy and 0.139 eV mean site-selection regret. Thus, within IMP2D, pair-held-out transfer almost entirely recombines familiar constituents and is substantially easier than transfer to a new constituent, while interstitial incorporation remains the more difficult energy-prediction regime.

I. INTRODUCTION

Point defects and incorporated impurity atoms can alter carrier density, localized electronic states, magnetism, and catalytic activity in two-dimensional (2D) materials. Their formation energies compare a relaxed defective structure with a pristine host and an atomic reservoir, and therefore depend on both the local defect environment and the surrounding chemical reference state [1]. Defect structures and energetics can differ between bulk and monolayer hosts [2], while the accessible surface and internal voids introduce distinct adsorbate and interstitial incorporation geometries [3]. Systematic first-principles surveys remain demanding because the search space grows across host chemistry, impurity identity, incorporation class, and symmetry-distinct site. Once such a survey exists, a separate question is whether learned representations can organize its energy landscape and expose where transfer across chemistry or incorporation geometry fails.

The IMP2D database provides a systematic first-principles survey of neutral adsorbate and interstitial impurities in 2D hosts [3]. Its scale makes supervised energy estimation possible, and graph neural networks are natural models because they operate directly on periodic atomic structures. ALIGNN demonstrated the value of joint bond and angle representations for crystalline properties [4], while SchNet established continuous-filter message passing from interatomic distances [5]. Defect-specific learning has developed along several complementary directions. Bulk defect energetics have been treated with defect graph networks, semiconductor graph mod-

els, persistent-homology features, and charge-aware predictors [6–9]. For 2D systems, sparse defect representations have been evaluated on the 2DMD data, whereas tree and descriptor models have been applied directly to IMP2D [10–12]. These studies establish descriptor baselines for IMP2D and show that representation and data preparation materially affect reported errors. They also motivate architectures that expose the impurity site while retaining local coordination and longer-range periodic context. A random structure split, however, can place closely related host–impurity chemistries or numerically equivalent structures on both sides of the train–test boundary, leading to optimistic estimates [13]. It also conflates two physically different questions: combining a known host with a known impurity in a previously unseen pair, and predicting a host or impurity absent from all target supervision. Materials models are known to degrade when test chemistry differs from training chemistry [14, 15], but this distinction has not been resolved for impurity incorporation energies in IMP2D. The role of incorporation geometry—adsorbate versus interstitial—is likewise hidden by one pooled error.

Here we develop the Defect-Aware Radial Transformer (DART), an impurity-centered graph model that combines defect-local message passing, radial-bias self-attention, and graph readout for a non-extensive formation-energy target [Fig. 1]. A paired 2^3 factorial quantifies module effects, after which grouped tests distinguish transfer to unseen host–impurity pairs from transfer to hosts or impurities absent from IMP2D target supervision. Retrospective reranking metrics are computed from pair-held-out predictions for existing DFT-

relaxed candidates. Because those geometries and their reference energies come from the same completed DFT workflow, this analysis does not replace structure generation or relaxation. A separate ensemble experiment assesses internal uncertainty calibration and selective prediction [16–18].

II. METHODS

A. Dataset, target provenance, and canonical modeling set

We use neutral impurity structures from the Impurities in Two-Dimensional Materials Database (IMP2D) [3]. The source ASE database contains 17,364 rows. Replaying the data-preparation filter in database order removes 4551 unconverged calculations, 1828 rows with a missing or nonfinite stored formation energy, and 344 rows with $|E_f| > 20$ eV, leaving 10,641 structures. These structures span 44 hosts, 65 impurity elements, and two incorporation classes (adsorbate and interstitial).

The two incorporation geometries are illustrated schematically in Fig. 2.

For rows with complete source fields, the target was independently recomputed as

$$E_f = E_{\text{def}} - E_{\text{host}} - \mu_{\text{imp}}. \quad (1)$$

Here E_{def} and E_{host} are the total DFT energies of the relaxed impurity-containing and pristine host supercells, respectively. The IMP2D convention takes μ_{imp} as the per-atom energy of the impurity element in its relaxed crystalline standard state, evaluated with the source workflow [3]. For site or incorporation-class comparisons within one host–impurity pair, E_{host} and μ_{imp} are common offsets and cancel. The released relaxed impurity geometry and E_{def} are products of the same completed IMP2D DFT workflow. Consequently, supplying that geometry does not avoid the relaxation that produced the reference energy. DART receives the relaxed impurity-containing structure rather than the three terms of Eq. (1); it learns their difference implicitly for retrospective energy estimation and reranking, conditional on the available geometry and the IMP2D reference convention. The recomputed and stored values agree to a maximum absolute numerical difference of 2.3×10^{-13} eV. Four otherwise filtered rows contain a zero sentinel in the stored defective-structure energy and therefore cannot be independently reconstructed from the source components; these rows are excluded from every modeling split.

The released relaxed structures do not retain a persistent identity field for the atom initially inserted by the structure generator. In 349 rows, the impurity element is also a constituent of the host, so multiple identical nuclei match the impurity label. Of these cases, 316 are interstitial and 33 are adsorbate structures. Assigning one of them by array order would break permutation invariance

and would make the defect-centered features depend on a nonphysical convention. We therefore require exactly one atom to match the impurity element and exclude all 349 non-unique rows from the protocol.

We also prevent detected numerical equivalents from crossing split boundaries. Duplicate candidates are identified by the union of two fingerprints: (i) atomic numbers, cell, and Cartesian coordinates rounded to 10^{-6} , and (ii) a permutation- and translation-invariant spectrum of element-labeled periodic pair distances rounded to 10^{-4} Å. The latter detects numerically equivalent structures even when atom order or coordinate origin differs. Among the remaining eligible structures, connected duplicate groups contain 64 redundant rows; the lowest-index auditable representative is retained. The resulting modeling set contains 10,224 structures. The complete audit, excluded sample indices, database identifiers, and split files are distributed with the code.

B. Fixed evaluation partitions

All partitions were generated once from the curated set and stored as index lists. Five paired random 80/10/10 partitions (assignment seeds 42–46) are used for the 2^3 architecture experiment. A separate fivefold random cross-validation partition provides one out-of-fold prediction for every retained sample. Chemical transfer is evaluated using balanced fivefold group cross-validation in which complete host identities, impurity identities, or host–impurity pairs are held together. In each grouped fold, one group fold is used for testing and the next group fold for validation. We further note that pair grouping does not mathematically require each marginal host and impurity to remain in training. A split audit found one singleton-host test structure for which this condition fails; the other 10,223 pair-held-out test structures retain both constituent identities in their corresponding training fold. We further use a predefined matrix-completion holdout: group-VI dichalcogenide hosts combined with 3d impurities form the test block, while group-IV/V hosts combined with 4d impurities form the validation block. After the common audit exclusions, this partition contains 9,487 training, 372 validation, and 365 test structures.

Uncertainty quantification uses a separate random partition with 75% for training, 10% for checkpoint selection, 5% for calibration, and 10% for internal testing (assignment seed 62). Calibration and test indices are excluded from ensemble training and checkpoint selection, and test labels are not used for calibration. All split files cover the full 10,641-row container through mutually disjoint train, validation, test, optional calibration, and exclusion sets.

The sequential data audit, retained chemical coverage, target distribution, and fixed partition geometry are summarized in Supplemental Fig. S1 [19].

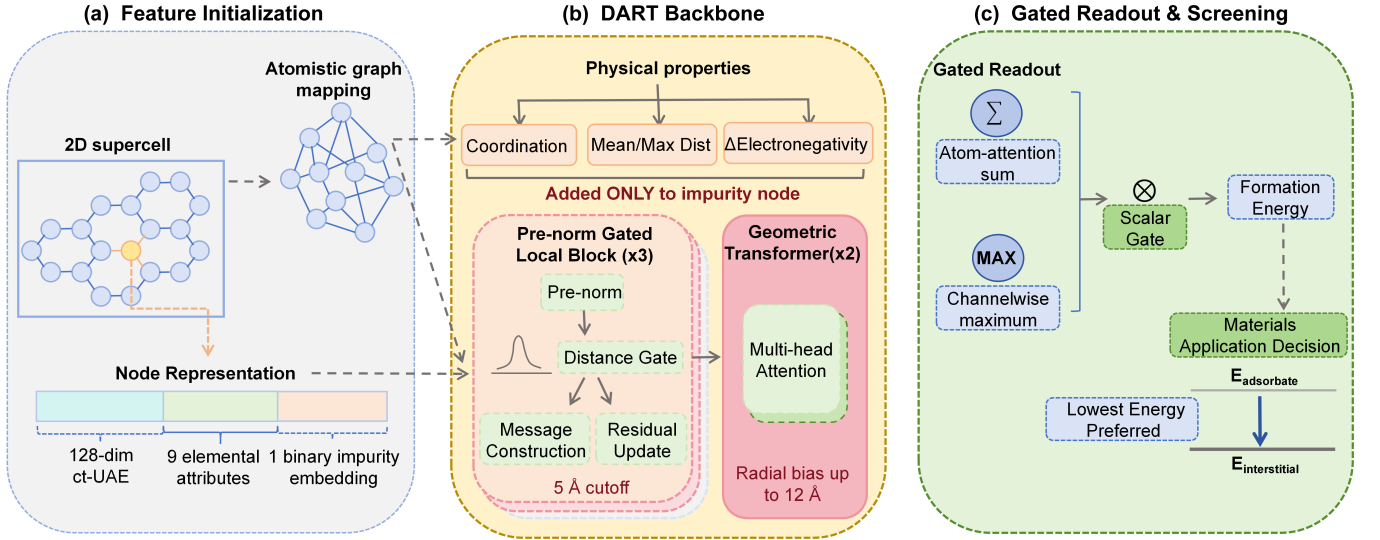


FIG. 1. Architecture of the Defect-Aware Radial Transformer (DART). (a) Feature initialization of the atomistic graph. A fixed 128-dimensional ct-UAE vector and nine normalized elemental attributes are concatenated and projected from 137 to 128 channels, after which a learned binary impurity-status embedding marks the unique impurity node. (b) The DART backbone adds four projected defect-environment descriptors only at that node before three 5-Å pre-normalized, distance-gated local blocks and two four-head geometric Transformer blocks. The radial bias uses 32 basis functions spanning 0–12 Å, while attention connects all valid atoms. (c) A graph-dependent scalar gate fuses an atom-attention weighted sum and a channelwise maximum before a $128 \rightarrow 128 \rightarrow 1$ head predicts the formation energy of each structure. For each host-impurity pair, retrospective incorporation-class reranking compares the minimum predicted energy among retained adsorbate candidates with the corresponding minimum among interstitial candidates; the displayed ordering is illustrative.

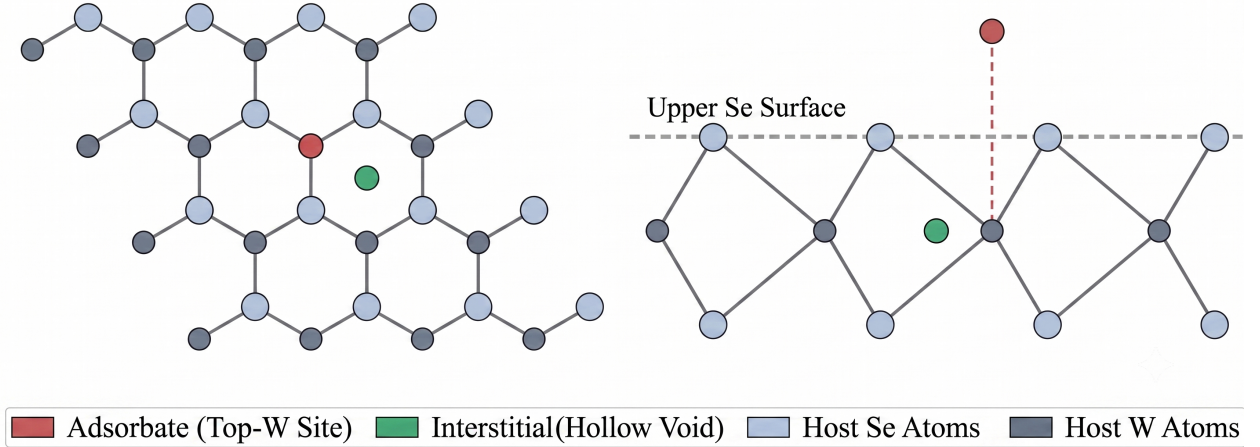


FIG. 2. Structural representation of the two neutral impurity incorporation classes in two-dimensional host materials. The overlaid top and side views show two alternative single-impurity configurations in a WSe_2 -like host: a red adsorbate at a top-host site (Top-W in this example) and a green interstitial impurity in a hollow void. The alternatives do not coexist in a single structure and illustrate representative, rather than exhaustive, IMP2D incorporation sites.

C. Defect-aware graph Transformer

Each periodic supercell is represented by an atomistic graph. We refer to the resulting model as the Defect-Aware Radial Transformer (DART). DART uses the defect environment as a local inductive bias while retaining global host context. A uniquely identified impurity node exposes the defect center, short-range message passing

represents its coordination environment, periodic radial attention supplies longer-range response, and graph-level readout maps the combined representation to one formation energy. The initial node representation concatenates a fixed 128-dimensional ct-UAE elemental vector with nine elemental attributes: group, period, Pauling electronegativity, covalent and van der Waals radii, a group-based valence count, first ionization energy, elec-

tron affinity, and atomic mass [20]. The nine columns are min–max normalized over atomic numbers 1–100. An unavailable van der Waals radius is replaced by the corresponding covalent radius; the other tabulated values, including explicit unavailable or approximate entries, are fixed by the released feature table. We reconstruct a 100-element table from the published checkpoint by applying its linear `atom_embed` layer to each one-hot elemental code, $\mathbf{U} = \mathbf{W}^\top + \mathbf{1}\mathbf{b}^\top$. After a zero-padding row is prepended, atomic number Z indexes table row Z , which corresponds to row $Z - 1$ of $\mathbf{U}$. The concatenated representation is projected to 128 hidden channels, and a learned binary embedding marks the impurity atom. Three local interaction layers aggregate neighbors within 5 Å. Distances are expanded in 32 Gaussian radial basis functions, and three-body messages use a 32-function angular expansion. Two subsequent four-head geometric Transformer blocks apply self-attention to all valid atoms, with a learned radial bias constructed from 32 basis functions centered between 0 and 12 Å [21]. A two-layer multilayer perceptron maps the pooled graph representation to $\hat{E}_f$. Because every retained row passes the identity audit, the binary impurity embedding always marks exactly one compositionally unique atom; no array-order heuristic is used. Figure 1 summarizes this data path, from the 5-Å local block through the radial-biased geometric Transformer to gated graph readout.

The confirmatory architecture experiment varies three binary modules while holding every other training and model setting fixed:

Gated readout (G): A learned atom-attention sum and a channelwise maximum are fused by a graph-dependent scalar gate. The off state uses masked mean pooling.

Environment enrichment (E): Four defect-centered quantities—coordination number, mean and maximum neighbor distance, and the absolute electronegativity contrast to the local neighborhood—are normalized, projected, and added only to the impurity representation. The off state omits this branch.

Pre-normalized, distance-gated local block (P):

The on state normalizes node states before message construction and applies a learned scalar distance gate to each edge before a residual update. The off state uses the original ungated message and post-residual normalization [22]. We treat this as one composite local-block change and do not attribute its effect to normalization or edge gating separately.

For the selected local block, let $\tilde{\mathbf{h}}_i$ be the layer-normalized state of atom i , $\mathbf{r}(d_{ij})$ its radial basis, and g_{ij} a learned scalar gate. Pair and angle messages are

$$\mathbf{m}_{ij} = \phi_r[\mathbf{r}(d_{ij})] \odot W_v \tilde{\mathbf{h}}_j g_{ij}, \quad g_{ij} = \sigma\{\phi_g[\mathbf{r}(d_{ij})]\}, \quad (2)$$

$$\mathbf{t}_{jik} = \phi_\theta[\tilde{\mathbf{h}}_i \|\mathbf{a}(\theta_{jik})], \quad (3)$$

where $\mathbf{a}$ is the angular basis and $\|$ denotes concatenation. The graph builder retains at most the first 32 ordered neighbor pairs at each center; denote that retained set by $\mathcal{T}_i$. The residual update is

$$\mathbf{h}'_i = \mathbf{h}_i + \phi_n \left(\sum_j \mathbf{m}_{ij} + \sum_{(j,k) \in \mathcal{T}_i} \mathbf{t}_{jik} \right). \quad (4)$$

For attention head a , the subsequent global block uses

$$A_{ij}^{(a)} = \text{softmax}_j \left[\frac{\mathbf{q}_i^{(a)} \cdot \mathbf{k}_j^{(a)}}{\sqrt{d_h}} + b_a \{\mathbf{r}_g(D_{ij})\} \right], \quad (5)$$

where D_{ij} is the norm obtained by wrapping each fractional-coordinate difference independently into $[-1/2, 1/2)$ and mapping it through the cell. This componentwise wrapping is periodic but is not guaranteed to select the shortest image in an oblique cell. The 32 global radial basis centers span 0–12 Å; beyond the outer center their activations decay toward zero, while attention still connects every valid atom. The gated readout is

$$\mathbf{h}_{\text{att}} = \sum_i \alpha_i \mathbf{h}_i, \quad (6)$$

$$\mathbf{h}_{\text{max}} = \max_i \mathbf{h}_i, \quad (7)$$

$$s = \sigma\{\phi_s(\mathbf{h}_{\text{att}} \|\mathbf{h}_{\text{max}})\}, \quad (8)$$

$$\mathbf{h}_G = \text{LN}[s\mathbf{h}_{\text{att}} + (1 - s)\mathbf{h}_{\text{max}}]. \quad (9)$$

Elemental scalars, local distances and angles, and wrapped global displacement norms make the scalar prediction invariant to rigid translation and rotation of a periodic structure. Pair aggregation, attention, and graph pooling are symmetric, but the capped triplet enumeration follows the stored neighbor order; exact invariance to atom-index permutations is therefore not claimed.

Figure 3 illustrates the four local quantities entering the E branch and their association with the uniquely identified impurity node.

The full 2^3 factorial contains all eight G/E/P combinations on each of the five paired random partitions. Within a partition, all variants share the same initialization seed and training batches. Main effects and interactions are orthogonal factorial contrasts. The architecture used in subsequent transfer and uncertainty experiments is the variant with the lowest mean validation MAE; exact ties are resolved by fewer enabled modules and then by binary variant code. Test metrics are not part of the ordering rule.

D. Training protocol

Networks are optimized for 150 epochs with AdamW, a peak learning rate of 5×10^{-4} reached after a ten-epoch linear warmup from 5×10^{-5} , weight decay 10^{-4} , and cosine decay to 10^{-6} . The objective is mean absolute

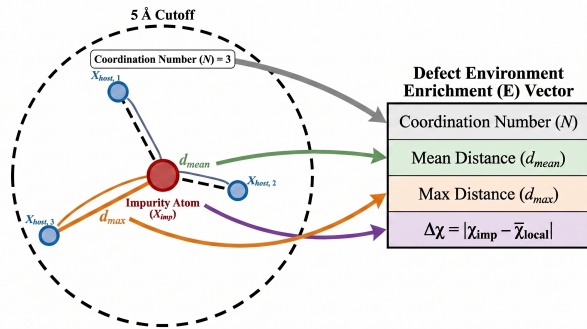


FIG. 3. Illustration of the defect-environment enrichment (E) module. Within the periodic 5-Å neighbor graph, four impurity-centered quantities—the neighbor count N , mean and maximum neighbor distances (d_{mean} and d_{max}), and the normalized Pauling-electronegativity contrast $\Delta\chi = |\chi_{\text{imp}} - \langle\chi_{\text{nn}}\rangle|$ —form a four-component descriptor. The descriptor is normalized, projected to 128 channels, and added only to the impurity-node representation before local message passing. The $N = 3$ neighborhood shown is schematic.

error (MAE), batch size is 64 for DART, and gradient norms are clipped at 5. Targets are standardized using training-partition statistics for optimization and transformed back to eV for all reported errors. The fixed DART regularization recipe is limited to 0.1 dropout and additive Gaussian target noise with standard deviation 0.03 eV; no coordinate or strain augmentation is applied. A host-balanced sampler draws 200 structures per host per epoch; its sequence and the independent target-noise stream are deterministically indexed by model seed and epoch so paired variants see the same batches and noise realizations. Every factorial variant uses the same two input assets. The ct-UAE table described above is a non-trainable buffer. Separately, we pretrained the nine-feature input projection and three ungated, post-normalized local interaction layers for 30 epochs. We first drew 20,000 records uniformly without replacement from JARVIS-DFT `dft_3d` with seed 42; conversion and graph-validity filters retained 19,902 pristine three-dimensional crystals [23]. Pretraining used formation energy per atom as the source target, an 80/10/10 random partition, and seed 42. No IMP2D target is used in this stage. An explicit chemistry-overlap audit found that 26 of the 44 IMP2D host labels (25 of 42 distinct reduced host formulas) occur among 69 JARVIS source records, and that all 65 IMP2D impurity elements occur somewhere in the source-task structures. Formula overlap does not establish structural identity, but it means that the grouped partitions withhold IMP2D formation-energy supervision rather than all prior source-task exposure to the same reduced host formula or impurity element. Accordingly, “host held out” and “impurity held out” denote absence from IMP2D target supervision, not absence from the JARVIS source-task chemistry.

The checkpoint initializes the nine-feature slice of the input projection, its bias, and all compatible parameters

of those three local-interaction layers. The ct-UAE slice and the P-specific edge gates retain their seeded initialization. All trainable parameters are then optimized on the assigned IMP2D partition. We select the epoch with minimum validation MAE and evaluate it once on the corresponding test partition. Before aggregation, partition membership is checked and all regression metrics are recomputed from the stored targets and predictions.

E. Baselines

Two baseline classes use exactly the same partitions. First, an 80-dimensional deterministic descriptor uses atomic number, mass, covalent radius, and Pauling electronegativity. Mean, standard deviation, minimum, maximum, and median over all atoms and host atoms contribute 40 entries; impurity values and impurity-minus-host-mean contrasts contribute eight. Eighteen entries encode atom count, elemental diversity, cell lengths, area, volume, mass density, the first six impurity-neighbor distances, two coordination counts, and local-distance mean and standard deviation. Twelve site indicators and two incorporation-class indicators complete the vector. Missing electronegativities use the fixed value 2.0 and fewer than six neighbors are padded at 20 Å. We fit a training-target mean predictor, ridge regression, random forests, histogram gradient boosting, and LightGBM [24]. These model families also permit direct comparison with earlier IMP2D descriptor studies [11, 12]. Hyperparameters within each family are selected by validation MAE only. The learned descriptor family is then chosen independently inside every split by that split’s validation MAE; pooled scores concatenate the corresponding fold-wise selected test predictions. Thus no held-out fold’s test labels enter the family used to predict that fold. The training-target mean predictor is reported separately and is not a candidate in family selection.

Second, the predefined additive-readout periodic SchNet recipe [5] uses 128 hidden channels, six interaction blocks, 50 Gaussian functions, and a 5 Å cutoff. Its graph prediction sums learned atomwise scalar contributions, following the additive energy readout of the original SchNet formulation. It uses batch size 32 but otherwise follows the same target normalization, 150-epoch AdamW schedule, host balancing, gradient clipping, and target-noise protocol as DART; it uses a conventional learned atomic-number embedding and no DART pre-training. Random and host-impurity-pair fivefold evaluation uses one fixed model seed per fold (242 for DART and 342 for SchNet). Host-held-out, impurity-held-out, and chemistry-block evaluation uses three independently initialized seeds (242–244 for DART and 342–344 for SchNet). Predictions are averaged within each split before pooled out-of-fold metrics are calculated. Consequently, DART–SchNet contrasts compare the complete learning recipes, including their different input representations, initialization, and graph readouts; they do not

isolate a pure backbone-architecture effect. The selected DART has 820,558 trainable parameters, compared with 455,937 for SchNet.

To assess the unusually large host-held-out SchNet error, we repeated that evaluation after replacing only atomwise addition with mean pooling. All other model, optimization, split, and seed settings were unchanged. Predictions were averaged over the three seeds within each fold, and the mean-minus-add difference in absolute error was assigned a 95% percentile interval from 20,000 host-cluster bootstrap samples. This post hoc sensitivity is reported separately from the main benchmark.

F. Internal uncertainty calibration and retrospective reranking

Five independently initialized members of the validation-selected DART architecture are trained on the dedicated uncertainty split, following the deep-ensemble construction and its atomistic use as a model-credibility signal [16, 25]. Ensemble mean and sample standard deviation provide the point prediction and a raw model-disagreement proxy. A fixed-seed random permutation (seed 6201) assigns the dedicated calibration rows to two subsets without using target labels. One subset fits a nonnegative scale and floor for

$$\sigma_i = \sqrt{(as_i)^2 + b^2} \quad (10)$$

by Gaussian negative log likelihood; the other subset estimates finite-sample split-conformal quantiles of $|E_{f,i} - \hat{E}_{f,i}|/\sigma_i$ [17, 26]. Under exchangeability within this split, the resulting intervals have finite-sample marginal coverage at each evaluated level; the guarantee is not conditional on host or impurity identity. Because architecture development used other partitions of the same corpus, we interpret the reported coverage as internal calibration rather than as an untouched outer-holdout estimate. Test labels are not used in calibration.

For retrospective reranking, the five host-impurity-pair folds yield one out-of-fold prediction per retained structure. Within each host-impurity pair we compare the minimum predicted and reference energies of adsorbate and interstitial configurations, identify the predicted lowest-energy site, and report preference accuracy and reranking regret. We define the incorporation-class margin as

$$\Delta E_{\text{cls}} = E_{\text{min}}^{\text{int}} - E_{\text{min}}^{\text{ads}}, \quad (11)$$

so positive values favor adsorption. Reranking regret is the excess reference energy of the configuration selected by the prediction over the reference minimum in the eligible candidate set. Exact-site metrics require at least two candidate sites, whereas top-two accuracy requires at least three; reference minima within 10^{-8} eV are treated as tied, and tied adsorbate-interstitial pairs are excluded from binary preference accuracy.

Accuracy is contextualized by two non-trained descriptive decision references defined over the eligible evaluation decisions. Incorporation class uses the empirical majority class among eligible IMP2D labels. Exact-site accuracy uses the analytic expectation from choosing uniformly among the observed candidates, and top-two accuracy uses the corresponding uniformly sampled size-two subset; tied reference minima count as successful choices. Model-minus-reference intervals use the same host-impurity-pair cluster bootstrap as the model estimates. These are predictive associations within IMP2D, not causal mechanism assignments or external DFT validation.

III. EVALUATION PROTOCOL

A. Metrics and statistical analysis

We report MAE, root-mean-square error (RMSE), signed bias, Spearman rank correlation, and R^2 . Host- and impurity-macro MAE give each chemical group equal weight, independent of its number of structures. Cross-validation predictions are concatenated only when test folds are disjoint; model seeds are averaged within a fold before concatenation. Low-energy behavior is assessed by MAE for $E_f \leq 0$, MAE within the true lowest-energy decile of each pooled test regime, and recall of that decile by the predicted lowest-energy decile. These strata are evaluation summaries and are not used for architecture or hyperparameter selection.

For the architecture factorial, each effect is evaluated as the difference between the average metric at positive and negative levels of the associated orthogonal contrast within each paired repeat. Ninety-five-percent intervals are obtained by nonparametric bootstrap over the five repeat-level contrasts. Because only five paired repeats were used, the intervals are accompanied by repeat-level sign consistency and interpreted as descriptive uncertainty summaries rather than large-sample hypothesis tests. For model comparisons on aligned out-of-fold predictions, we use 10,000 paired bootstrap resamples. The resampling unit matches the held-out unit: individual structures for random cross-validation, complete hosts for host-held-out evaluation, complete impurities for impurity-held-out evaluation, and host-impurity pairs for pair-held-out and chemistry-block evaluation. Within each resampled chemical unit all associated structures remain together. Reranking intervals use 20,000 host-impurity-pair cluster-bootstrap resamples. Adsorbate and interstitial site decisions from the same pair remain together, while point estimates average all eligible decision sets. An interval excluding zero is treated as evidence for a directional difference; otherwise the result is reported as inconclusive rather than equivalent.

Uncertainty evaluation includes empirical interval coverage and mean width at 50%, 80%, 90%, and 95% nominal coverage, Gaussian negative log likelihood, the contin-

uous ranked probability score, Spearman correlation between uncertainty and absolute error, and risk-coverage curves. The area under the risk-coverage curve (AURC) is accompanied by its excess over oracle ordering [27]. All calibration choices use only the dedicated calibration partition.

All energy estimates and reranking metrics are retrospective against the retained IMP2D targets and conditioned on geometries produced by the same DFT workflow as the reference energies. The study does not evaluate an independently generated, target-matched first-principles test set or claim to replace candidate generation and relaxation.

IV. RESULTS

A. Effects of defect-aware model components

Validation MAE selected the all-enabled **g111** variant. Paired contrasts resolved lower errors for gated readout and the composite pre-normalized, distance-gated local block, whereas the independent main effect of environment enrichment was inconclusive. The complete factorial is reported in Table I, with its paired-effect visualization in Supplemental Fig. S2 [19]. The selected variant had mean validation MAE 0.371 eV (95% bootstrap interval [0.357, 0.385] eV) across the five paired repeats and test MAE 0.382 eV [0.377, 0.387] eV.

The orthogonal contrasts separate evidence for the three modules from variation between random partitions. Because *P* jointly changes normalization placement and distance gating, its contrast supports or rejects that composite block only; it does not identify either subchange as the causal source of an effect. On validation MAE, the enabled-minus-disabled main-effect contrasts were *G*: $\Delta = -0.018$ [-0.020, -0.016] eV (supported reduction); *E*: $\Delta = -0.003$ [-0.010, +0.004] eV (inconclusive); *P*: $\Delta = -0.022$ [-0.029, -0.018] eV (supported reduction). Directional validation intervals were also observed for *G* : *P*: $\Delta = -0.007$ [-0.011, -0.003] eV (supported reduction); *E* : *P*: $\Delta = +0.0048$ [+0.0001, +0.0098] eV (supported increase). All other prespecified interaction intervals included zero. The repeat-level contrast agreed with the pooled point-estimate direction in *G* 5/5, *P* 5/5, *G* : *P* 5/5, *E* : *P* 4/5 paired repeats. With five paired repeats, the percentile intervals and sign counts are descriptive uncertainty summaries rather than large-sample hypothesis tests.

B. Interpolation, chemical transfer, and low-energy behavior

Chemical transfer was strongly axis dependent. Mean absolute error was 0.450 eV when complete host-impurity pairs were withheld, versus 0.418 eV under random cross-validation. Withholding complete hosts or impurities in-

creased MAE to 0.938 eV and 0.844 eV, respectively. The predefined chemistry block had MAE 0.386 eV but samples a different target distribution and should therefore be interpreted separately rather than placed on a monotonic extrapolation scale. Across all five regimes, paired absolute-error differences favored DART over both the descriptor model selected by validation MAE within each split and the additive-readout SchNet baseline. Figure 4 places the five transfer regimes and their paired comparator contrasts on a common visual scale; Table II reports the complete intervals.

Aggregate MAE can hide failures in chemically sparse or reranking-relevant subsets. Table III therefore reports equal-weight host and impurity macro errors together with performance on nonpositive formation energies and the lowest-energy decile. For example, on pair-held-out predictions, the true low-decile MAE was 0.480 eV and its recall by the predicted low decile was 86.0%. None of these low-energy summaries entered architecture or hyperparameter selection.

Incorporation geometry introduced an additional asymmetry. Under pair-held-out prediction, interstitial structures had MAE 0.719 eV, compared with 0.322 eV for adsorbates. Host-, impurity-, and site-resolved errors were also heterogeneous; the complete descriptive analysis is reported in Supplemental Fig. S4 [19].

C. Internal uncertainty and selective prediction

On the internal ensemble test partition, the five-member ensemble had MAE 0.377 eV, Gaussian negative log likelihood 1.128, and mean Gaussian CRPS 0.324 eV. The Spearman correlation between calibrated standard deviation and absolute error was 0.546. The risk-coverage curve had AURC 0.177 eV and excess AURC 0.095 eV relative to oracle error ordering.

At 90% nominal coverage, the split-conformal interval covered 87.2% of test structures with mean width 1.141 eV. Supplemental Table SII and Fig. S3 [19] report all evaluated nominal levels, the risk-coverage curve, and the error-uncertainty relation. These values measure empirical marginal coverage within the internal ensemble split, not calibration under a chemically shifted distribution.

D. Retrospective out-of-fold reranking within IMP2D

The host-impurity-pair folds provide one out-of-fold prediction for every retained structure. Both the input geometries and reference energies come from the completed IMP2D DFT workflow, so the following comparisons are retrospective rather than a substitute for relaxation. Across pairs containing both incorporation classes, DART identified the lower-energy class among the retained, DFT-relaxed IMP2D candidates with 91.0% ac-

TABLE I. Paired 2^3 architecture factorial. Values are mean MAE with 95% bootstrap intervals over five paired repeats, in eV. G denotes gated readout, E defect-environment enrichment, and P the composite pre-norm distance-gated local block. Bold marks the validation-selected variant; test MAE was not used in its selection.

Variant	G	E	P	Validation MAE	Locked test MAE
g000	0	0	0	0.422 [0.413, 0.430]	0.419 [0.410, 0.433]
g001	0	0	1	0.394 [0.375, 0.413]	0.402 [0.393, 0.413]
g010	0	1	0	0.404 [0.390, 0.417]	0.412 [0.405, 0.421]
g011	0	1	1	0.401 [0.396, 0.407]	0.410 [0.394, 0.429]
g100	1	0	0	0.401 [0.389, 0.413]	0.413 [0.407, 0.420]
g101	1	0	1	0.374 [0.357, 0.388]	0.387 [0.375, 0.400]
g110	1	1	0	0.404 [0.396, 0.410]	0.405 [0.394, 0.415]
g111	1	1	1	0.371 [0.357, 0.385]	0.382 [0.377, 0.387]

TABLE II. Pooled out-of-fold or single-block test MAE (eV). SchNet-add denotes the prespecified periodic SchNet recipe with atomwise-additive readout. The descriptor column selects a learned family independently within every split by that split’s validation MAE (Histogram GB, LightGBM were selected across the resulting folds). Δ is comparator minus DART absolute error with a paired 95% bootstrap interval using the regime-matched resampling unit; positive values favor DART.

Regime	DART	SchNet-add	Descriptor	$\Delta_{\text{SchNet-add}}$	$\Delta_{\text{desc.}}$
Random OOF	0.418	0.571	0.625	+0.154 [+0.136, +0.174]	+0.208 [+0.194, +0.221]
Pair OOF	0.450	0.628	0.674	+0.178 [+0.157, +0.202]	+0.224 [+0.207, +0.242]
Host OOF	0.938	6.703	1.266	+5.765 [+3.006, +9.296]	+0.328 [+0.205, +0.450]
Impurity OOF	0.844	1.692	1.336	+0.848 [+0.626, +1.092]	+0.492 [+0.368, +0.623]
Chemistry block	0.386	0.508	0.642	+0.122 [+0.069, +0.175]	+0.256 [+0.189, +0.323]

curacy; the predicted and reference class margins had Spearman correlation 0.954 and margin MAE 0.749 eV. Selecting one configuration globally gave 69.5% exact-site accuracy and mean regret 0.139 eV. When the incorporation class was fixed, exact and top-two site accuracies were 73.6% and 91.4%, respectively, with mean regret 0.103 eV. The corresponding descriptive reference accuracies were 66.2%, 23.4%, 37.1%, and 62.1%, respectively. Relative to these references, DART improved accuracy by 24.8 [22.4, 27.2], 46.1 [44.0, 48.3], 36.5 [35.0, 38.0], and 29.2 [28.0, 30.4] percentage points, respectively.

Complete decision metrics and clustered intervals are reported in Supplemental Table SIII [19]. Figure 5 shows the class-margin agreement, the distribution of reranking regret, and the clustered decision-accuracy intervals.

V. DISCUSSION

A. Local and global context shape model choice

The paired factorial separates module effects from variation between data partitions. Validation contrasts supported small reductions in MAE for gated readout (G) and the composite local block (P), but did not resolve an independent effect of environment enrichment (E). The negative G:P and positive E:P interactions indicate non-additive module effects and help account for selection of the all-enabled configuration even though E lacks a resolved main effect. The four E descriptors may overlap with information already encoded by local distance and

angle messages, but the factorial does not establish that they are redundant. Likewise, P changes both normalization placement and edge gating, so its contrast applies only to the composite block.

Graph readout provides a direct connection between the prediction target and model design. Atomwise addition is appropriate for extensive total energies, whereas the IMP2D formation energy is a defect excess energy and is non-extensive with respect to host atom count. Local coordination is a useful inductive bias, but the energy difference can retain long-range elastic or electronic contributions; DART therefore also includes all-atom attention. Pooling choice is known to affect atomistic prediction of intensive and localized quantum properties [28]. Replacing only the additive SchNet readout by mean pooling reduced its host-held-out MAE from 6.703 eV to 2.552 eV and improved all five folds. For this evaluated SchNet recipe, the controlled sensitivity identifies additive pooling as an important contributor to its large host-held-out error; it does not establish a universal pooling rule for formation energies. The remaining gap to DART (0.938 eV) reflects the complete learning recipes, including their inputs, initialization, interactions, and readouts; it is not evidence for a backbone-only advantage. Fold-resolved errors and size-error diagnostics are reported in the Supplemental Material [19].

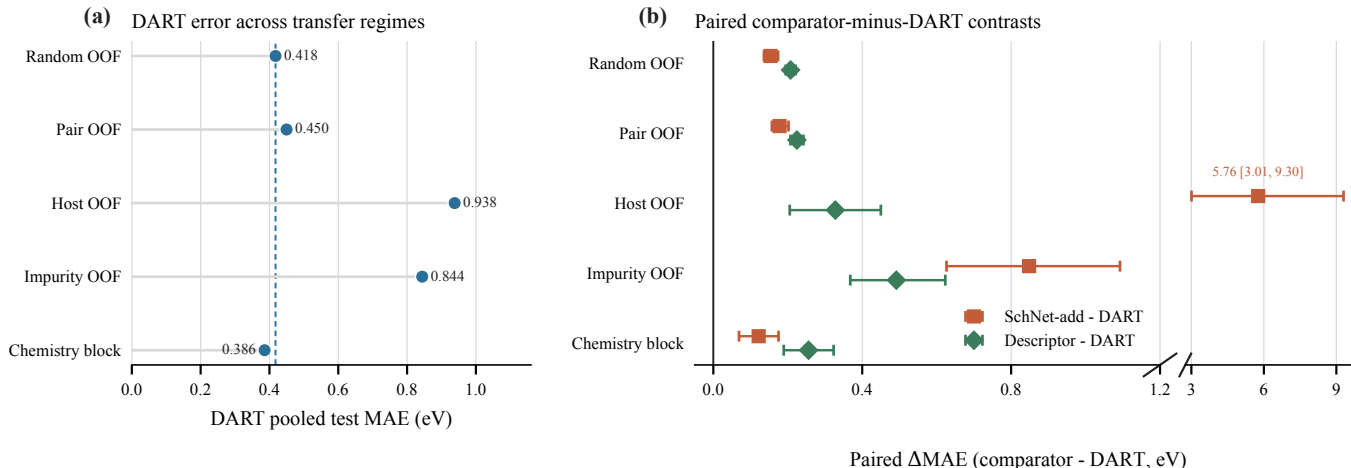


FIG. 4. Interpolation and chemical-transfer evaluation. (a) DART pooled test MAE across the five evaluation regimes; the dashed line marks random out-of-fold MAE. (b) Paired differences in absolute error for the additive-readout SchNet baseline and the descriptor model selected separately by validation MAE in each split, relative to DART. Intervals are 95% bootstrap intervals using the regime-matched resampling units defined in Sec. III A; the broken axis isolates the large host-held-out SchNet contrast. Positive values favor DART.

TABLE III. Applicability-domain diagnostics for DART. Macro MAEs weight each host or impurity equally. Nonpositive-target MAE uses $E_f \leq 0$; low-decile MAE and recall use the true and predicted lowest 10% of each pooled test regime. Energy errors are in eV and recall is in percent. A dash denotes a regime with no eligible nonpositive test targets.

Regime	Host macro	Impurity macro	Nonpositive-target MAE	Low-decile MAE	Low-decile recall
Random OOF	0.489	0.440	0.396	0.440	87.1
Pair OOF	0.533	0.468	0.450	0.480	86.0
Host OOF	0.990	0.968	1.134	1.284	69.2
Impurity OOF	0.887	0.826	0.973	1.041	71.6
Chemistry block	0.384	0.387	—	0.305	75.0

B. Chemical axes and incorporation geometry define applicability

The difference between random and chemistry-grouped evaluation is physically structured. Pair-held-out MAE was 0.450 eV, compared with 0.418 eV under random cross-validation. In the pair-held-out case, all but one test structure had both its host and impurity represented elsewhere in training, so this regime almost entirely tests recombination of familiar constituent representations. Withholding every IMP2D target for a host or impurity instead increased MAE to 0.938 eV and 0.844 eV. This separation shows that pair-held-out recombination is a less severe transfer problem than an unseen constituent within this database.

The grouped tests do not represent element- or formula-disjoint pretraining. The JARVIS source task contains reduced-formula matches for 26 IMP2D host labels and contains every IMP2D impurity element. Host- and impurity-held-out therefore mean absence from IMP2D formation-energy supervision, not absence from all earlier representation learning. The results remain internal to the 44 hosts and 65 impurities retained from IMP2D and should not be extrapolated to arbitrary

2D compounds or reference conventions.

Incorporation geometry creates a second hierarchy. Interstitial MAE under pair-held-out prediction was 0.719 eV, more than twice the 0.322 eV adsorbate error. Interstitial sites are typically more highly coordinated and geometrically confined than adsorbate sites in the source data [3], which offers a plausible source of greater structural sensitivity, but the present analysis does not isolate a mechanism. The retained population is also selective: the impurity-identity audit removes 316 interstitial and 33 adsorbate structures whose inserted atom cannot be identified permutation-invariantly. The observed class asymmetry therefore applies to the retained population rather than to every released IMP2D row.

C. Retrospective energy estimation and reranking of available relaxed geometries

For a fixed host-impurity pair, the pristine-host energy and impurity chemical potential in Eq. (1) cancel when sites or incorporation classes are ranked. This cancellation helps explain why useful relative decisions can coexist with a larger absolute formation-energy error. Pair-

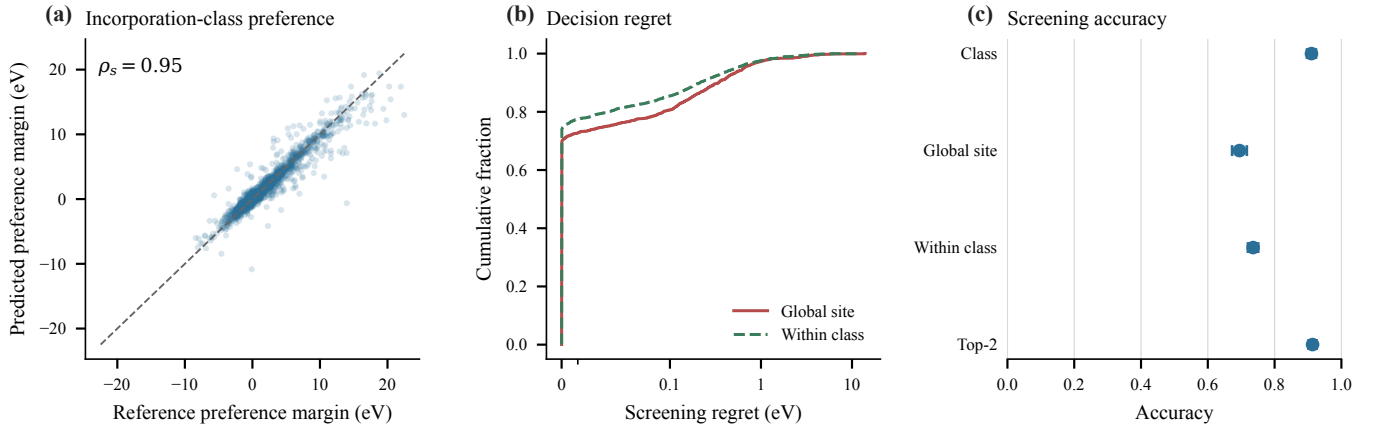


FIG. 5. Retrospective reranking of available DFT-relaxed IMP2D geometries using pair-held-out predictions. (a) Predicted versus reference preference margins between adsorbate and interstitial configurations. (b) Empirical cumulative distributions of global and within-class reranking regret. (c) Host-impurity-pair cluster-bootstrap intervals for incorporation-class, exact-site, and top-two decision accuracy.

held-out predictions identified the lower-energy incorporation class with 91.0% accuracy, and their mean global site-selection regret was 0.139 eV. Exact-site accuracy measures recovery of the reference minimum, whereas regret distinguishes a serious miss from selection of a nearly degenerate alternative. The top-two result further quantifies whether a small follow-up queue retains the reference minimum.

These decisions are conditional on the DFT-relaxed candidate geometries released by IMP2D. Because each geometry and its reference energy are outputs of the same completed DFT workflow, the model does not bypass the relaxation cost that produced the target. The analysis therefore concerns retrospective energy estimation and reranking over available relaxed candidates, not structural generation, relaxation, thermodynamic populations, kinetics, or synthesizability.

D. Internal uncertainty and remaining scope

On the dedicated ensemble split, nominal 90% intervals had 87.2% empirical coverage and mean width 1.141 eV. Coverage and width must be interpreted together, and the nonzero oracle gap in Supplemental Fig. S3 [19] shows that ensemble disagreement does not recover all high-error cases. Because the calibration and test rows come from the same finite IMP2D population, these results support internal marginal calibration and risk ranking, not host-conditional, impurity-conditional, or chemically shifted coverage.

The evidence is limited to neutral adsorbates and interstitials, the source database’s relaxed structures and reference energies, and a model that infers the reference-energy difference from the impurity-containing graph alone. Vacancies, substitutions, charged defects, transition levels, migration barriers, finite-temperature free

energies, and competing phases are outside scope. Duplicate control removes numerical equivalents detected by the two declared structural fingerprints but cannot prove that every physical redundancy has been found. SchNet and the descriptor families provide aligned comparators rather than an exhaustive survey of atomistic architectures.

Two implementation details also limit geometric interpretation. The angle branch retains the first 32 ordered neighbor pairs at each center, making that branch sensitive to stored atom order when more pairs are available. The global radial bias wraps fractional-coordinate differences componentwise, which need not select the shortest image in an oblique cell. The reported results are reproducible for the released ordering and implementation, but strict permutation robustness and a true shortest-image global bias have not been established.

A target-matched external set would provide the most direct next test: its structures, chemical potentials, convergence settings, and target definition would need to match the IMP2D convention. A complementary extension would calibrate uncertainty directly on host- or impurity-shifted partitions, where exchangeability is intentionally stressed.

VI. CONCLUSION

Impurity formation-energy transfer within IMP2D is organized by chemical novelty. Pair-held-out folds, with one singleton-host exception, gave 0.450 eV MAE, compared with 0.418 eV under random cross-validation. Removing all target supervision for a host or impurity increased the error to 0.938 eV and 0.844 eV, respectively. Incorporation geometry added a second axis: interstitial structures were substantially harder to predict than adsorbates in the pair-held-out population.

The architecture results connect graph pooling to a formation-energy target that is non-extensive in host atom count. Gated graph readout and the composite pre-normalized, distance-gated local block produced small, repeat-consistent gains, whereas explicit environment enrichment had no resolved independent effect. In the evaluated SchNet recipe, a post hoc sensitivity test implicated atomwise addition, which is natural for an extensive total energy, as one contributor to the large host-held-out error.

For existing DFT-relaxed candidate geometries, pair-held-out predictions selected the lower-energy incorporation class with 91.0% accuracy and 0.139 eV mean global site-selection regret. Because the geometries and reference energies arise from the same completed DFT workflow, these results support retrospective energy estimation and reranking within the neutral, retained IMP2D domain rather than a reduction in relaxation cost. Testing newly generated structures prepared independently of their reference energies, together with uncertainty calibration under explicit chemical shift, remains necessary before prospective use beyond that domain.

ACKNOWLEDGMENTS

This work was supported by the National College Students Innovation and Entrepreneurship Training Pro-

gram (Key Support Area Project). Computations were performed on an NVIDIA L40S cluster at Wuhan University. OpenAI Codex (GPT-5, accessed August 2026) assisted with implementation and debugging of analysis and verification scripts, manuscript restructuring, language revision, and consistency checks. The authors directed these uses, reviewed the AI-assisted code and analyses, and verified the manuscript against the archived results, source code, and cited literature. No AI tool generated or modified the IMP2D reference labels, and the authors take responsibility for the final manuscript.

DATA AVAILABILITY

The source data are available as version 2 of the Interstitial and Adsorbate Structure Database [29]; their construction is described in Ref. [3]. The sample audit, split definitions, resolved configurations, analysis code, and machine-readable results used in this study are available in the project repository [30].

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